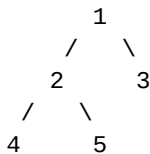


Tree Data Structure in C++

What is a Tree?

A Tree is a non-linear data structure used to store data in hierarchical form. It starts from a root node and every node can have child nodes.

Binary Tree Diagram



Tree Node Structure (C++)

```
// Structure of a tree node
struct Node {
    int data;           // Data stored in node
    Node* left;         // Pointer to left child
    Node* right;        // Pointer to right child
};
```

Create Node Function

```
// Function to create a new node
Node* createNode(int value) {
    Node* newNode = new Node(); // Allocate memory
    newNode->data = value;       // Assign data
    newNode->left = NULL;        // Initialize left child
    newNode->right = NULL;      // Initialize right child
    return newNode;
}
```

Complete Binary Tree Program

```
#include <iostream>
using namespace std;

// Node structure
struct Node {
    int data;
    Node* left;
    Node* right;
};

// Create a new node
Node* createNode(int value) {
    Node* newNode = new Node();
```

```

        newNode->data = value;
        newNode->left = NULL;
        newNode->right = NULL;
        return newNode;
    }

    // Inorder Traversal: Left -> Root -> Right
    void inorder(Node* root) {
        if (root == NULL)
            return;
        inorder(root->left);
        cout << root->data << " ";
        inorder(root->right);
    }

    int main() {
        // Creating tree
        Node* root = createNode(1);
        root->left = createNode(2);
        root->right = createNode(3);
        root->left->left = createNode(4);
        root->left->right = createNode(5);

        cout << "Inorder Traversal: ";
        inorder(root);

        return 0;
    }

```

Tree Traversals

1. Inorder Traversal: Left $\rightarrow$ Root $\rightarrow$ Right
2. Preorder Traversal: Root $\rightarrow$ Left $\rightarrow$ Right
3. Postorder Traversal: Left $\rightarrow$ Right $\rightarrow$ Root

Applications of Trees

Trees are used in file systems, databases, compilers, artificial intelligence, network routing, and searching algorithms.